

CS471 – Quiz 7 Study Guide

Lessons 26 & 27: ML Foundations & Naive Bayes Classification

Study Strategy

This study guide synthesizes the core mathematical concepts and tactical applications from Lessons 26 and 27. Focus on understanding *why* an algorithm is used and *how* it fails, rather than just memorizing the formulas.

Lesson 26: Machine Learning Foundations

Core Paradigms

- **Supervised Learning:** The algorithm is trained on a labeled dataset (an answer key). It learns to map inputs (features) to known outputs.
- **Unsupervised Learning:** The algorithm is given unlabeled data and must discover hidden structures or groupings on its own.
- **Generalization:** The ultimate goal of ML. A model must perform accurately on *entirely new, unseen data* (the Testing Set), not just memorize the data it studied (the Training Set).

The Bias-Variance Tradeoff

The expected error of any model is: $E = \text{Bias}^2 + \text{Variance} + \sigma^2$ (Irreducible Noise).

- **High Bias (Underfitting):** The model makes overly simplistic, rigid assumptions. It fails to capture the complex reality of the data.
- **High Variance (Overfitting):** The model perfectly memorizes the training data, including the random noise. It is highly sensitive to slight fluctuations and fails to generalize in combat.

Operational Metrics (The Confusion Matrix)

Flat accuracy is misleading if the target is rare (e.g., finding a single SAM site in a desert).

- **Precision** $\left(\frac{TP}{TP+FP}\right)$: Quality of the strike. When the model pulls the trigger, how often is it right? Maximize this when False Positives (collateral damage) are unacceptable.
- **Recall** $\left(\frac{TP}{TP+FN}\right)$: Completeness of the scan. Out of all actual threats, how many did it find? Maximize this when False Negatives (missed threats) are fatal.
- **F1-Score:** The harmonic mean of Precision and Recall. Provides a single, balanced summary of combat effectiveness.

Classification Algorithms

- **Logistic Regression:** Uses a linear equation but passes the output through a **Sigmoid function** to squash the value into a strict probability between 0.0 and 1.0.
- **Linear Discriminant Analysis (LDA):** Draws straight decision boundaries. It makes the strict assumption that all classes share the *exact same variance* and covariance shape.
- **Quadratic Discriminant Analysis (QDA):** Drops the LDA assumption. It allows every class to possess its own unique variance shape, allowing the mathematical boundaries to curve around specific, non-linear flight envelopes.
- **Support Vector Machines (SVM):** Seeks the optimal hyperplane that separates classes with the maximum possible margin (buffer zone). Data points on the edge of the margin are **Support Vectors**. Uses the **Kernel Trick** to project complex 2D data into 3D space to slice it with a flat plane.

Decision Trees & Random Forests

- **Decision Tree:** Splits data by asking Yes/No questions to maximize node purity.
- **Gini Impurity:** Measures purity. A Gini score of 0.0 means the node is perfectly pure (contains only one class). A single tree grown to a Gini of 0 will massively **overfit** (High Variance).
- **Random Forest:** Fixes high variance using **Bagging (Bootstrap Aggregating)**. It trains hundreds of shallow trees on random subsets of data and features, then averages their votes.

Lesson 27: Naive Bayes Classification

Bayes' Theorem

Bayes' Theorem formalizes how we update our beliefs based on new evidence.

$$P(\text{Class} \mid \text{Evidence}) = \frac{P(\text{Evidence} \mid \text{Class}) \times P(\text{Class})}{P(\text{Evidence})}$$

- **Prior $P(\text{Class})$:** The baseline historical probability before seeing any new evidence.
- **Likelihood $P(\text{Evidence} \mid \text{Class})$:** If the class is true, how likely is this specific evidence?
- **Posterior $P(\text{Class} \mid \text{Evidence})$:** The final, updated probability conclusion.

Text Classification & The "Naive" Assumption

- **Bag of Words:** To classify text rapidly, Naive Bayes ignores grammar, punctuation, and word order, treating the sentence as an unstructured collection of evidence.
- **The "Naive" Assumption:** The algorithm makes the massive mathematical assumption that **every single piece of evidence is completely independent**.
- **The Tactical Flaw:** In reality, evidence is correlated (e.g., the words "password" and "reset" often appear together). Because Naive Bayes assumes they are independent, it multiplies them as separate events, resulting in **double-counting the evidence** and producing mathematically overconfident scores.

The Zero-Frequency Problem & Laplace Smoothing

Naive Bayes calculates the final score by multiplying the probabilities of all words together.

- **The Absolute Veto:** If an intercepted message contains a brand new word the model has never seen, its historical probability is 0. Because anything multiplied by zero is zero, this single word acts as an absolute veto, dragging the entire threat score to zero regardless of how suspicious the rest of the message is.
- **Laplace (Add-One) Smoothing:** Fixes the absolute veto. It mathematically adds "1" to the count of every single word in the vocabulary, pretending the model has seen every possible word at least once. This turns the 0 into a tiny fraction, allowing the math to survive and the rest of the sentence to drive the classification.

Key Equations Reference

1. **Precision:** $\frac{TP}{TP+FP}$

2. **Recall:** $\frac{TP}{TP+FN}$

3. **F1-Score:** $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

4. **Gini Impurity:** $1 - \sum p_i^2$

5. **Laplace Smoothing Likelihood:** $P(w \mid C) = \frac{\text{count}(w,C) + \alpha}{\text{total words in } C + \alpha \times |V|}$